



Review

Synthetic therapeutic gene circuits in mammalian cells

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ARTICLE INFO

Article history:

Received 25 April 2014

Accepted 5 May 2014

Available online 17 May 2014

Edited by Wilhelm Just

Keywords:

Cell therapy

Gene circuits

Gene regulation

Gene therapy

Synthetic biology

ABSTRACT

In the emerging field of synthetic biology, scientists are focusing on designing and creating functional devices, systems, and organisms with novel functions by engineering and assembling standardised biological building blocks. The progress of synthetic biology has significantly advanced the design of functional gene networks that can reprogram metabolic activities in mammalian cells and provide new therapeutic opportunities for future gene- and cell-based therapies. In this review, we describe the most recent advances in synthetic mammalian gene networks designed for biomedical applications, including how these synthetic therapeutic gene circuits can be assembled to control signalling networks and applied to treat metabolic disorders, cancer, and immune diseases. We conclude by discussing the various challenges and future prospects of using synthetic mammalian gene networks for disease therapy.

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1. Introduction

The rapid development of molecular biology and bioinformatics, has enabled the rational, systematic design and creation of biological networks with desired functionality that do not exist in nature [1–3]. This emerging field is referred to as synthetic biology, which is defined as the science of the design and construction of novel functional devices, systems, and even organisms by applying engineering and computational principles [4,5]. In the relatively early stages of synthetic biology, many complex synthetic gene regulatory circuits were designed and created in prokaryotes or lower eukaryotes, including toggle switches [6], oscillators [7–9], timers [10], counters [11], clocks [12], pattern detectors [13], band-pass filters [14], and intercellular communication systems [15,16]. The development of these pioneering devices has given rise to various therapeutic possibilities for synthetic biology. Examples include designing a bacteriophage to switch off the bacterial SOS DNA repair system [17], engineering bacteria or viruses for cancer-targeting therapy [18–23], engineering *Escherichia coli* to prevent cholera infection [24], and engineering yeast cells to produce the precursor of the antimalarial drug artemisinic acid [25]. Although it is relatively simple to engineer synthetic therapeutic circuits in prokaryotic systems, the clinical applications of

these prokaryotic synthetic systems are limited. Therefore, increasing numbers of studies have focused on the design of mammalian transgene control devices that form the basis for the assembly of synthetic therapeutic networks in mammalian cells. Most of these control devices are designed to respond to small molecule chemicals [26–33]. Recently, more elegant circuits were developed to use traceless inducers such as light and radio waves to regulate transgene expression [34–36]. Furthermore, in the latest studies, several gene networks were created to monitor pathological levels of metabolites in the body, such as uric acid, fatty acids, bile acids, and dopamine [31,32,37,38]. These pioneering studies have opened a door for the development of new strategies to treat human diseases (Fig. 1). This review highlights the latest state-of-the-art synthetic gene circuits and prototype treatment strategies for therapeutic purposes.

2. Synthetic gene circuits for the treatment of metabolic disorders

2.1. Combating diabetes

Type 2 diabetes is a rapidly rising global epidemic characterised by islet beta-cell dysfunction, insulin resistance, and high blood glucose level [39,40]. It is a worldwide medical condition causing mortality in both developing and developed countries. The currently available therapeutic strategies include daily insulin

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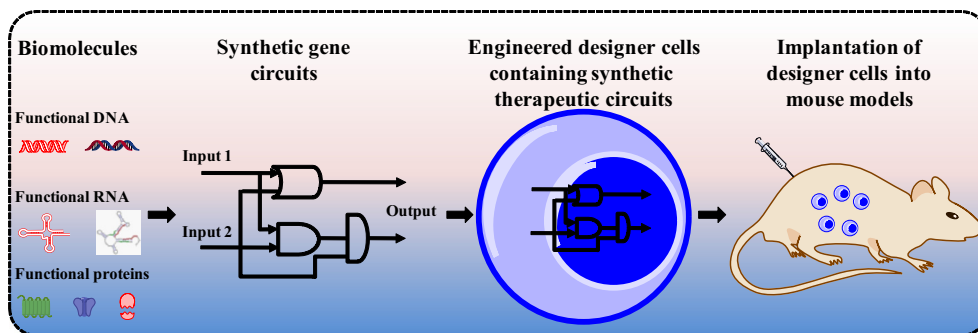


Fig. 1. The engineering principles of designed gene circuits for disease therapy in mammalian synthetic biology. Synthetic biologists capitalise on naturally existing biomolecules, including functional DNA, RNA, and proteins from a range of organisms such as microbes, plants, and mammals. These valuable biomolecules have been biochemically studied and are well understood. By using engineering principles, synthetic biologists design and reassemble these standard basic biobricks in a rational way to engineer state-of-the-art therapeutic gene circuits, which are then engineered into microencapsulated mammalian cell implants. These engineered cell implants can be transplanted into mouse models of disease and are regulated to produce therapeutic proteins to treat diseases.

injections, taking medicine before every meal, and strict dietary control [41].

Recently, two promising synthetic biology-based therapeutic strategies were developed for the treatment of diabetes. The first case is the design and construction of an optogenetic gene circuit to maintain blood glucose homeostasis [34]. This pioneering study by Ye et al. [34] used blue light as a traceless, molecule-free trigger signal to regulate transgene expression in animals. In this design, melanopsin is ectopically expressed in HEK-293 cells and used as a light receptor. Upon illumination, melanopsin signalling leads to elevated intracellular calcium levels and the cellular response to this is rewired to promote calcium-mediated activation of the transcription factor NFAT (nuclear factor of activated T cells). This signalling cascade leads to the activation of the P_{NFAT} promoter and expression of the glucagon-like peptide 1 (GLP-1), a peptide hormone that can restore blood glucose homeostasis in type 2 diabetes. Engineered cells containing this synthetic circuit were encapsulated in immunoprotective microcontainers and implanted into a type 2 diabetic mouse model. Upon blue-light illumination, the treated animals exhibited improved blood-glucose homeostasis (Fig. 2a). The utilisation of light as a trigger signal to regulate insulin expression for the treatment of diabetes was independently confirmed by another study [42].

Another study performed by Stanley et al. used radio waves as a trigger signal to control insulin expression and maintain blood glucose homeostasis in diabetic mice [35]. In this system, a temperature-sensitive receptor, TRPV1, is antibody-coated with iron-oxide nanoparticles and triggers a calcium surge when heated by radio waves. Calcium signalling is rewired to promote calcium-mediated activation of insulin expression. Mice with transplanted tumour xenografts containing this synthetic circuit were exposed to radio waves, which activated the expression and release of insulin from the tumours and, therefore, lowered blood glucose levels in the diabetic mice [35].

2.2. Treating the metabolic syndrome

Metabolic syndrome is a primary, epidemic public health challenge in the 21st century that is characterised by the three interlinked and interdependent metabolic disorders of hypertension, hyperglycemia, and obesity [43,44]. The currently available therapeutic strategy is to identify and treat each metabolic disorder and risk factor independently [41]. Until now, there has been no well-coordinated therapeutic strategy.

Recently, Ye et al. developed a multifunctional synthetic gene circuit for simultaneously treating multiple risk factors of metabolic syndrome [45] (Fig. 2b). The combination of drug and gene

therapies may improve the chances of clinical success. Guanabenz is a clinically approved antihypertensive drug that shows agonistic effects on the trace amine receptor 1 cTAAR1, whose activation triggers a strong cAMP second messenger response in mammalian cells. Therefore, the pharmacodynamics of guanabenz was functionally rewired via the signal transduction pathway of cTAAR1 to initiate transgene expression through the activation of CREB1 mediated by a cAMP-dependent phosphokinase A (PKA). Transcription initiation is driven by a synthetic promoter containing CREB1-specific, cAMP-response elements. Thus, this synthetic gene network enables guanabenz to dose-dependently control the expression of a fusion hormone GLP-1-Fc-Leptin, which simultaneously attenuates hypertension (guanabenz), hyperglycaemia (GLP-1), and obesity (Leptin) in mice with metabolic syndrome.

2.3. Maintaining uric acid homeostasis

Uric acid is the end-product of purine metabolism in mammals, and its homeostasis can be disrupted by many factors, leading to hyperuricemia and associated pathologies including the tumour lysis syndrome and gout [46,47].

A synthetic biology-based prosthetic network was designed to sense and monitor host metabolic parameters and respond accordingly to reverse the risk factors. Kemmer et al. developed a type of sensor-effector prosthetic network to sense the uric acid concentrations in the blood and control the expression of a secretion-engineered urate oxidase (smUOX) accordingly, which can restore urate homeostasis [31] (Fig. 2c). In this network design, smUOX expression dynamics are regulated by a bacterial HucR repressor, which is fused to the transcriptional-silencing domain KRAB (KRAB-HucR). Upon binding to its cognate operator, KRAB-HucR inhibits smUOX expression. However, when uric acid binds to HucR-KRAB and disrupts the binding of HucR-KRAB to its operator, smUOX is derepressed and secreted into the bloodstream where it converts urate into allantoin that is eliminated by renal clearance. Engineered cells containing this synthetic device were implanted into a mouse model of acute hyperuricemia. This synthetic device was able to stimulate the production of smUOX to dissolve uric acid crystal deposits in the kidneys of the mice and restore blood urate homeostasis [31].

2.4. Treating obesity

Diet-induced obesity (DIO) is a lifestyle-associated medical condition that results from excessive food energy intake, lack of physical activity, and genetic predisposition [48]. Obesity can increase the risk of developing a wide range of associated diseases, such

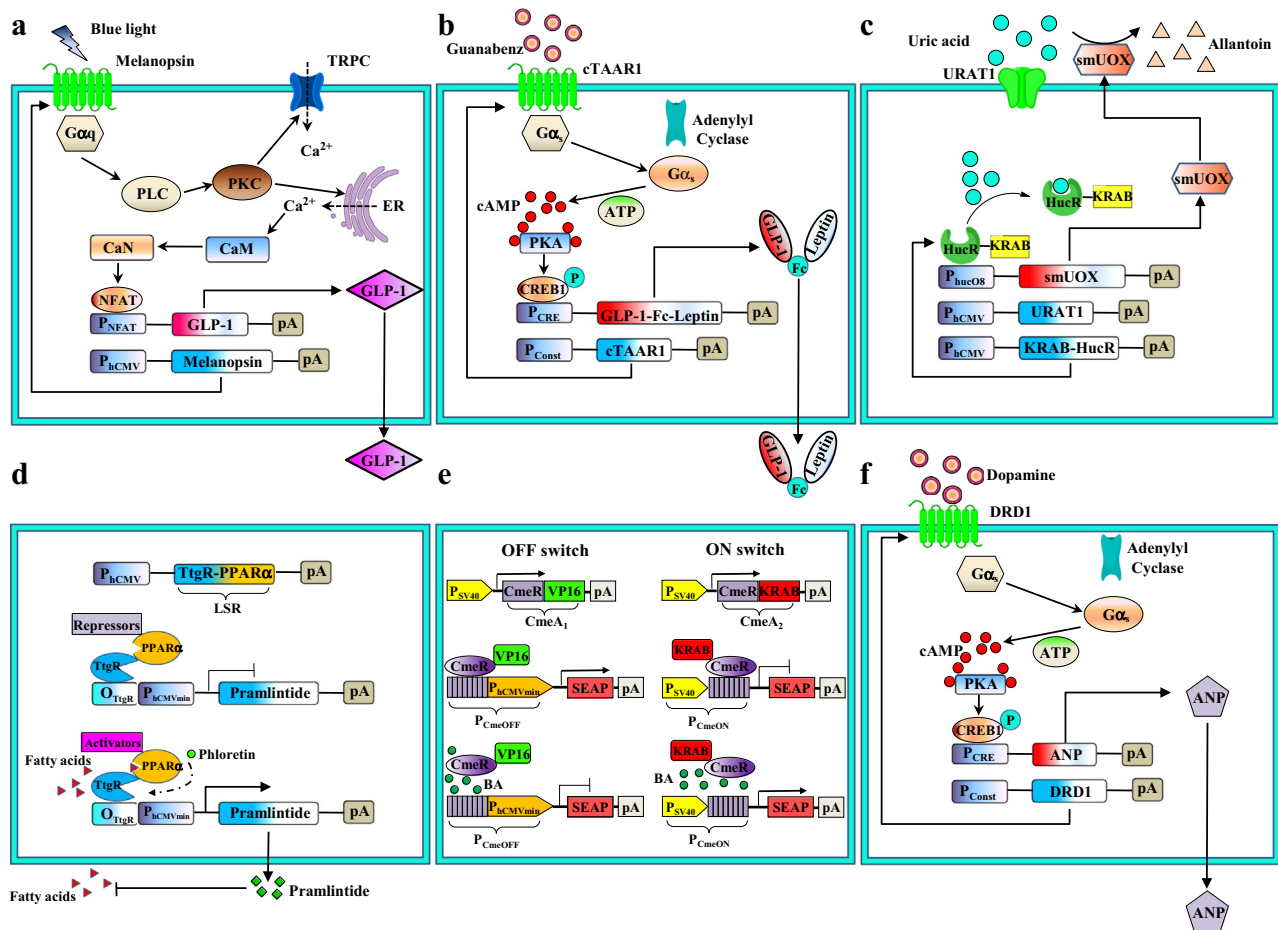


Fig. 2. Synthetic gene circuits designed for disease therapy. (a) A synthetic optogenetic transcription device for controlling blood-glucose homeostasis in diabetic mice. Upon blue-light illumination the ectopically expressed G-protein-coupled receptor (GPCR) melanopsin is activated and sequentially activates phospholipase C (PLC) through Gαq, and phosphokinase (PKC), which then triggers calcium influx through the activation of transient receptor potential channels (TRPCs) and from the endoplasmic reticulum (ER). The elevated intracellular calcium levels activate calmodulin (CaM) and then calcineurin (CaN), which in turn activates the transcription factor NFAT (nuclear factor of activated T cells) by dephosphorylation. The activated NFAT translocates into the nucleus, binds to its specific promoter (P_{NFAT}), and initiates expression of glucagon-like peptide-1 (GLP-1). The engineered HEK-293 cells incorporating this synthetic gene circuit were microencapsulated and subcutaneously implanted into diabetic mice. Upon external illumination of the mice by blue light, the synthetic gene circuit was activated and lowered the blood-glucose levels in the diabetic mice. (b) A pharmacaceutically controlled designer circuit for the treatment of metabolic syndrome. Binding of guanabenz to the ectopically expressed chimeric G-protein-coupled trace-amine-associated receptor (cTAAR1) activates adenylyl cyclase through Gαs, which converts ATP into cAMP. The intracellular cAMP surge is rewired to activate the cAMP-dependent phosphokinase A (PKA), which in turn activates the cAMP-response element-binding protein 1 (CREB1). Activated CREB1 binds to its synthetic promoter (P_{CRE}) and drives expression of the chimeric hormone protein GLP-1-Fc-Leptin. Mice with metabolic syndrome that were treated with guanabenz and implanted with the designer cells containing this synthetic circuit showed attenuated hypertension (guanabenz), hyperglycaemia (GLP-1), as well as dyslipidaemia and obesity (both Leptin). (c) Self-sufficient synthetic circuits control blood-urate homeostasis in mice. The heterologous expression of a urate transporter is able to increase the intracellular urate levels in designer cells. A uric acid sensor KRAB-HucR is designed to detect urate levels and can inhibit the transcription of the secreted urate oxidase variant (smUOX) by binding to its synthetic promoter (P_{hucO8}). When the urate concentration reaches pathological levels, the uric acid binds to KRAB-HucR, which is then released from P_{hucO8} and eventually triggers the expression of smUOX. smUOX lowers the blood-urate levels by converting urate into allantoin. (d) A closed-loop synthetic fatty-acid-controlling mammalian gene circuit for correcting obesity in mice. This synthetic fatty acid-responsive transcription circuit consists of a hybrid LSR, a fusion protein combining the phloretin-triggered repressor (TgR) and the ligand-binding domain of the human peroxisome proliferator-associated receptor alpha (PPARα), that can specifically bind to the TgR-specific operator linked to a minimal promoter (P_{hCMVmin}) to regulate transgene expression. In the absence of fatty acids, LSR associates with endogenous co-repressors to inhibit expression of the appetite-suppressing peptide hormone pramlintide. In the presence of fatty acids, LSR associates with endogenous co-activators to initiate pramlintide expression. The transgene expression can also be turned off by adding phloretin, which disrupts LSR-promoter binding. Mice with diet-induced obesity (DIO) implanted with microencapsulated designer cells containing this synthetic circuit demonstrated a significant reduction in food consumption, blood lipid levels, and body weight. (e) Synthetic bile acid-regulated OFF and ON switches. The OFF switch: the transcriptional repressor CmeR was fused to the *Herpes simplex*-derived transactivation domain (VP16) to obtain a bile acid-dependent transactivator, CmeA₁ (CmeR-VP16), driven by the simian virus 40 promoter (P_{SV40}). In the absence of bile acids (BA), CmeA₁ binds to its chimeric promoter P_{CmeOFF}, composed of a CmeR-specific operator module and a minimal version of the human cytomegalovirus immediate early promoter (P_{hCMVmin}), and activates SEAP expression. In the presence of BA, CmeA₁ dissociates from P_{CmeOFF} and the SEAP expression is switched off. The ON switch: CmeR is fused to the human Kruppel-associated box (KRAB) domain to obtain a BA-dependent transsilencer, CmeA₂ (CmeR-KRAB), that drives expression from the P_{CmeON} promoter. In the absence of BA, CmeA₂ binds to its specific tandem operator sites (O_{Cme}) and inhibits P_{SV40} driving SEAP expression. In the presence of BA, CmeA₂ is released from P_{CmeON} resulting in SEAP expression. (f) A synthetic reward-based dopamine-triggered signalling network for controlling blood pressure in hypertensive mice. The binding of dopamine to the heterologously expressed human dopamine receptor 1 (DRD1) activates adenylyl cyclase through Gαs, which catalyses ATP into cAMP. The intracellular cAMP surge is rewired to activate PKA, which in turn activates CREB1. Activated CREB1 binds to its synthetic promoter (P_{CRE}) and drives expression of the blood pressure-reducing atrial natriuretic peptide (ANP). When hypertensive mice were implanted with this synthetic sensor-effector device and their reward system was triggered by sexual arousal, the treated mice showed a significant increase in dopamine and blood ANP levels and decrease in blood pressure.

as cardiovascular disorders, type 2 diabetes, osteoarthritis, and certain types of cancer. It is becoming one of the most widespread epidemic diseases, affecting up to 1.5 billion people in developing as well as developed countries [49–51].

Recently, Roessger et al. designed a closed-loop genetic circuit that can constantly monitor blood fatty acid levels in DIO mice and express accordingly the clinically licensed appetite-suppressing peptide hormone pramlintide to reverse hyperlipidaemia and obesity in the animals [37] (Fig. 2d). In this synthetic gene circuit, pramlintide expression levels are controlled by the lipid-sensing receptor (LSR), a hybrid transcription factor, containing the ligand-binding domain of PPAR α and the bacterial DNA-binding repressor TtgR, showing dual-input sensitivity to fatty acids as well as the apple metabolite and cosmetic additive phloretin. In the absence of fatty acids, LSR recruits co-repressors and inhibits P_{TtgR}-driven transgene expression. In the presence of fatty acids co-activators bind to LSR, and the P_{TtgR} promoter starts to express pramlintide in a dose-dependent manner. However, in the presence of phloretin, LSR is released from P_{TtgR} and transcription is turned off irrespective of the lipid status of the cells. Hence, the TtgR component of the lipid-sensor ensures promoter specificity and also provides a potential safety switch to override lipid-dependent promoter activation. Engineered cells containing this synthetic circuit were implanted into wild-type mice fed with diets containing different amounts of fat. This resulted in diet-dependent expression of pramlintide, which suppresses appetite, promotes satiety, and slows gastric emptying, therefore limiting food intake, attenuating the hyperlipidaemic blood levels and restoring the body weight of treated mice [37].

Because obesity is associated with high serum levels of bile acids [52–54], Roessger et al. also designed and created a bile-acid-controlled transgene expression circuit in mammalian cells [32]. This synthetic metabolite-triggered gene circuit is also able to interface with the host metabolism and appropriately control transgene expression in mice (Fig. 2e). By fusing the bile-acid-dependent transcriptional repressor CmeR to different transactivation and trans-silencing domains, the authors created a bile-acid-sensitive OFF (BEAR_{OFF}) and ON switch (BEAR_{ON}). The BEAR_{OFF} circuit contains the bile-acid-dependent transactivator CmeA₁ (CmeR-VP16), which was assembled by fusing CmeR to the VP16 transactivation domain, and a cognate promoter (P_{CmeOFF}). HEK-293 cells cotransfected with the CmeA₁ expression vector and the CmeA₁-specific P_{CmeOFF}-driven SEAP reporter vector were treated with increasing concentrations of cholic acid and exhibited dose-dependent repression of SEAP expression. The BEAR_{ON} switch consists of the bile acid-dependent trans-silencer CmeA₂, which was assembled by fusing CmeR to the human Krueppel-associated box (KRAB) trans-silencing domain, and a CmeA₂-specific P_{CmeON} promoter. HEK-293 cells cotransfected with the CmeA₂ expression vector and the P_{CmeON}-driven SEAP reporter vector exhibited dose-dependent induction of SEAP expression. The BEAR_{ON} gene switch was further validated in mice, which verified that it could induce SEAP expression in animals treated with bile acids [32].

2.5. Controlling blood pressure

Dopamine is a key endogenous neurotransmitter that coordinates learning, controls emotions, and promotes reward- and pleasure-seeking behaviour [55–57]. Excessive reward-seeking behaviour may lead to addiction, drug abuse, schizophrenia, depression or obesity [58,59]. The pleasure status directly correlates with the dopamine levels released in the brain. Since dopamine leaks into bloodstream via the sympathetic nervous system, brain and blood dopamine levels are interrelated. Capitalizing on this correlation between brain and blood dopamine levels, Roessger et al. designed and created a synthetic dopamine-responsive

sensor–effector device that can monitor dopamine levels in the peripheral circulation and remotely control transgene expression in a dose-dependent manner [38]. This synthetic dopamine sensor–effector circuit can interface with reward-associated brain activities to regulate the clinically licensed atrial natriuretic peptide (ANP) for the treatment of hypertension. The dopamine sensor–effector circuit was designed to utilise ectopically expressed human dopamine receptor D1, whose native signalling cascade was rewired to induce cAMP-dependent protein kinase A (PKA)-mediated phosphorylation of cAMP response element-binding protein (CREB1), which can trigger ANP expression from synthetic promoters containing CREB1-specific operators (P_{CRE}). To confirm reward-triggered ANP expression *in vivo*, hypertensive mice implanted with microencapsulated HEK-293 cells transgenic for the dopamine-sensitive ANP expression were exposed to reward-triggering sexual arousal. Sexual arousal induced dopamine release and dopamine-dependent ANP expression, which reversed the hypertonic systolic blood pressure of the treated animals to normal levels (Fig. 2f).

3. Synthetic gene circuits for anti-cancer therapy

3.1. An AND logic-gated cancer kill switch

Cancer encompasses a large group of diseases characterised by uncontrolled cellular proliferation [60]. The abnormal cells can invade nearby healthy tissues and spread to other organs [60]. Cancer is one of the most important human diseases; therefore, a broad array of research has focused on finding new targets or therapies for the treatment of cancer. However, one of the key challenges for cancer therapy is to eliminate cancer cells without damaging the surrounding healthy tissues [60]. The most recent advances in cancer-targeting therapies include the engineering of oncolytic viruses [20,61], tumour-targeting bacteria [21–23], and synthetic cancer cell-specific kill switches [62–64].

Nissim et al. developed a dual-promoter integrator that can precisely discriminate and kill cancer cells [62]. Two different tumour-specific promoters drive the expression of two chimeric proteins (DocS-VP16 and Coh2-GAL4). Only when both of the two chimeric proteins are simultaneously expressed in response to cancer-indicating signals, do they dimerise to form a functional transactivator (GAL4-Coh2-DocS-VP16) that activates the P_{GAL4} promoter to trigger the expression of the *Herpes simplex*-derived thymidine kinase (HSV-TK). HSV-TK converts the prodrug ganciclovir to a toxic derivative, resulting in targeted cancer cell death (Fig. 3a). This synthetic dual-promoter circuit works as a digital-like AND logic gate that can improve targeting precision and efficacy in cancer cells.

3.2. A multi-input microRNA-based cancer classifier

In another study performed by Xie et al., a cancer-cell classifier was developed to identify HeLa-specific endogenous microRNA concentrations [63] (Fig. 3b). This multi-input classifier consists of a combination of genetic sensors that detect high and low levels of specific sets of microRNAs. If the expression of the specific input microRNAs matches a pre-defined high or low level profile, the cancer classifier drives expression of the pro-apoptotic hBax, which kills cancer cells such as HeLa. In the high microRNA sensor, high levels of specific microRNAs (microRNA-17 microRNA-21 and microRNA-30a) prevent the expression of the reverse tetracycline-dependent transactivator (rtTA) and the lactose operon repressor (LacI) in a two-level cascade which results in derepression of hBax transcription. In the low microRNA sensor, hBax expression is only permitted by low levels of a second set of specific microRNAs (microRNA-141, microRNA-142(3p), and

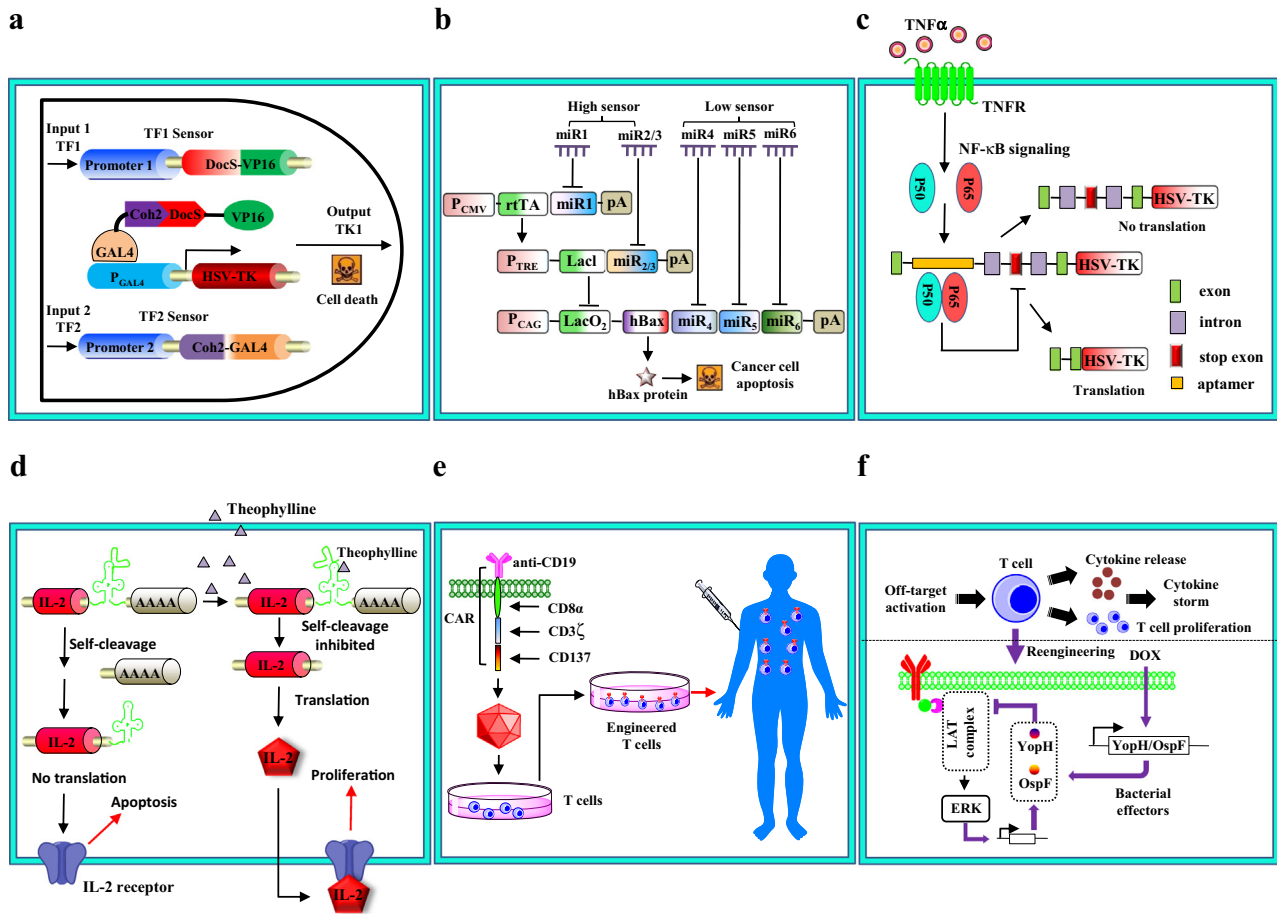


Fig. 3. Synthetic circuits for cancer and immune-mediated therapy. (a) Cancer-killing switch with AND expression logic. The cancer-killing switch contains two sensor units, which produce a cell-killing compound in the presence of two neoplastic markers. The two neoplastic markers act as transcription factors (input TF1, input TF2) which drive cognate promoters (promoter 1, promoter 2) and produce DocS-VP16 and Coh2-Gal4, respectively. Upon simultaneous expression, DocS-VP16 and Coh2-Gal4 heterodimerize to produce a synthetic transcription factor (Gal4-Coh2-DocS-VP16) which binds and activates a synthetic promoter (P_{GAL4}) and drives expression of the *Herpes simplex* virus type 1 thymidine kinase (HSV-TK). HSV-TK converts the prodrug ganciclovir into a toxic compound. (b) A multi-input microRNA (miR)-based cancer classifier. A HeLa cell-specific classifier monitors the levels of two sets of endogenous microRNAs and produces the pro-apoptotic cancer-killing gene hBax if the miRNA levels match a specific profile. In the miRNA high-sensor, high levels of microRNAs (miR1, miR2/3) prevent expression of the transcriptional cascade consisting of constitutive P_{CMV}-driven expression of the reverse tetracycline-dependent transactivator (rtTA) and rtTA-driven expression of the repressor of the lactose operon (LacI). Repression is mediated by binding of the miRs to complementary target sites in the rtTA- and lacI-encoding transcripts. In the absence of LacI, the cancer gene (hBax) is expressed by the constitutive P_{CAG} promoter. The miRNA low-sensor relies on miR4, miR5 and miR6 levels to be low, so they will not bind and interfere with P_{CAG}-driven hBax expression. As a consequence, hBax is exclusively expressed when miR1, miR2 and miR3 levels are high and miR4, miR5 and miR6 levels are low. (c) A synthetic protein-sensing RNA-based cancer-killing switch. The RNA-based synthetic cancer-kill switch is composed of p50/p60-specific aptamers that enable splice out of a stop codon from a synthetic three-exon-two-intron minigene placed in the 5' untranslated region of the *Herpes simplex* thymidine kinase (HSV-TK). The splicing event only occurs if tumor necrosis factor-α (TNFα) binds and activates the tumor necrosis factor receptor (TNFR), which activates the NF-κB signalling pathway and triggers nuclear translocation and binding of p50/p60 to the aptamer-containing minigene. Upon permissive splicing HSV-TK is expressed which kills the cell in the presence of the prodrug ganciclovir. (d) A synthetic RNA switch controlling T-cell proliferation. CTL-2 cells were reprogrammed to proliferate in response to the external trigger molecule theophylline. The proliferation of CTL-2 cells requires the presence of interleukin 2 (IL-2). The theophylline-dependent hammerhead ribozyme was engineered into the 3'-UTR of IL-2. The addition of theophylline inhibits the ribozyme self-cleavage and allows IL-2 translation which induces proliferation of the CTL-2 cells. (e) Synthetic signalling cascades for chronic lymphocytic leukemia treatment. The chimeric antigen receptor (CAR) contains four different domains: an anti-CD19 single chain fragment, a CD8α-derived transmembrane domain, a human CD137-derived co-stimulatory domain, and a CD3ζ signalling domain. Patient-derived T cells were transduced with CAR-encoding lentiviral particles and reinfused into the patient. CD19-positive cancer cells were recognized by CAR-transgenic T cells which resulted in expansion of the engineered T-cell population and killing of the tumor cells. (f) Controlling human T-cell response by rewiring kinase pathways with bacterial effectors. A synthetic amplitude limiter or a synthetic pause switch was designed to control cytokine release in isolated primary human CD4⁺ T cells. The synthetic pause switch was created by expressing YopH or OspF, which can inhibit specific steps in the T-cell receptor (TCR) signalling pathway. Doxycycline controls the expression of bacterial effectors (YopH and OspF), which are driven by a tetracycline-inducible promoter. The extracellular signal-regulated kinase (ERK), a key component of TCR signalling, could be inactivated by OspF, whereas the T-cell scaffold protein LAT could be dephosphorylated by YopH. Therefore, T-cell activation is controlled by externally added doxycycline.

microRNA-146a). Therefore, in the absence of microRNA-141, microRNA-142(3p) and microRNA-146a, and the presence of microRNA-17, microRNA-21, and microRNA-30a, this cancer classifier can induce apoptosis through production of the hBax protein. By integrating the intracellular levels of microRNAs, this synthetic logic decision-making circuit is able to specifically kill cancer cells [63].

3.3. A synthetic protein-sensing RNA-based cancer-killing switch

Culler et al. developed a synthetic protein-sensing RNA actuation device that autonomously detects and specifically kills cancer cells [64]. The device consists of a synthetic three-exon-two-intron minigene that contains aptamers specific for neoplastic markers (e.g., NF-κB or β-catenin) and controls functional splicing of the

linked *Herpes simplex*-derived thymidine kinase (HSV-TK). Binding of the neoplastic markers to the embedded aptamers of the HSV-TK-encoding transcript triggers specific splicing, expression of HSV-TK and results in HSV-TK-mediated cell-killing in the presence of the prodrug ganciclovir (Fig. 3c). This RNA switch only works as a cell-killing switch in the presence of endogenous cancer markers, while leaving the surrounding healthy tissue unharmed. The engineering of mammalian cells to perform logic calculations according to the presence or absence of specific cancer biomarkers is a promising option that may have great potential for more precise cell-based therapies in the future.

4. Synthetic gene circuits for immune-mediated therapy

4.1. A synthetic riboswitch controlling T-cell proliferation

T cells are a class of white blood cells that are essential to the mammalian immune system, especially the response to and defence against invasive pathogens and cancer cells [65–68]. Because human T cells can be easily isolated from patients, genetically engineered, and then returned into patients to treat cancer or chronic infection, they have great value for synthetic biologists as platforms for engineering therapeutic gene circuits [65,66,69]. One example is a synthetic RNA control switch constructed to externally control T-cell proliferation in mice [70] (Fig. 3d). In this study, the authors designed a synthetic ribozyme switch that enables programmable, drug-controlled expression of IL-2, a cytokine that is known to trigger proliferation of antigen-stimulated T cells. This was achieved by using a theophylline-responsive hammerhead ribozyme located in the 3'-UTR of the IL-2 gene. When engineered into T cells and reinfused into mice, the animals' T-cell population could be expanded by the administration of theophylline [70].

4.2. Engineering of T cells for tumour therapy

Another example is the reprogramming of T cells for tumour immunotherapy [69]. Porter et al. created a synthetic signalling cascade for chronic lymphocytic leukaemia therapy. In this study, the chimeric antigen receptor (CAR) was designed by assembling four different functional domains: an anti-CD19 single chain fragment, a CD8 α -derived transmembrane domain, a human CD137-derived co-stimulatory domain, and a CD3 ζ signalling domain. Primary human T cells were isolated from the patient and engineered to express the CAR. The engineered T cells were then returned into the patient. The patients' chronic lymphocyte leukaemia cells expressing the CD19 marker were recognised by the transgenic T cells expressing the chimeric antigen receptors. The transgenic T cells were activated to proliferate, and they killed the cancer cells [69] (Fig. 3e).

4.3. Synthetic pause switches for controlling T-cell overactivation

Engineering human T cells *ex vivo* and then transferring them back into patients to treat diseases is a promising strategy. However, the therapeutic application of engineered T cells for autoimmune disorders has the risk of adverse side effects including the inadvertent autoimmune-like attack of off-target host tissue [71,72]. It is necessary to control the specificity and amplitude of T-cell function. To balance therapeutic action and off-target toxicity, bacterial virulence proteins were recently explored as a tool to rewire kinase signalling pathways in human T cells [73] (Fig. 3f). Isolated primary human CD4⁺ T cells were reengineered to regulate cytokine release using an engineered synthetic amplitude limiter or a synthetic pause switch. The synthetic pause switch or the

synthetic negative feedback loop was created by expressing YopH or OspF, which can specifically inhibit the T-cell receptor (TCR) signalling pathway. A tetracycline-responsive promoter was engineered to control the expression of both bacterial effectors (OspF and YopH) by adding doxycycline. A central TCR signalling pathway, the MAPK extracellular signal-regulated kinase (ERK) pathway, could be inactivated by OspF, while YopH could specifically dephosphorylate a phosphotyrosine in the T-cell scaffold protein LAT. Therefore, T-cell activation could be externally controlled by adding doxycycline. In this study, the authors demonstrated the therapeutic applications of bacterial pathogen components as a tool to design and construct feedback modulators and external inducible pause switches, which can enable control of the T-cell response amplitude or temporarily disable T-cell activation [73].

4.4. Remote-controlling T-cell migration by a synthetic signal

As the key players in cell-based immunity T cells are highly motile [74–77]. Park et al. engineered T cells to specifically migrate up a gradient of clozapine-N-oxide (CNO), an inert metabolite of the FDA-approved antipsychotic drug clozapine. [78]. Such remote-control of cell mobility was achieved by engineering T cells for ectopic expression of the Di receptor, a designer receptor exclusively activated by a designer drug (DREADD), which rewires to conserved intracellular signalling and positive feedback loops that include the activation of the Rho-family GTPase Rac and the generation of phosphatidylinositol-(3,4,5)-Tris-phosphate by phosphatidylinositol 3-kinase (PI3K) at the leading edge of the migrating cells. Synthetic chemotaxis was confirmed in mice where engineered T cells preferentially migrated towards injected CNO-releasing beads [78]. Synthetic remote-control of T-cell migration may provide novel cell-based treatment strategies such as drug-mediated T-cell trafficking into tumours or to sites of infection.

5. Outlook

The engineering of gene circuits in mammalian cells has a wide range of applications and provides new strategies for the therapeutic application of synthetic biology. Recent work has led to the design of complex synthetic mammalian gene circuits assembled into biosensing devices that can monitor the metabolites within or surrounding cells and offer a promising strategy for biomedical treatment. These designed circuits are engineered to monitor, quantify, and treat a series of diseases by detecting disease-related specific markers and releasing a therapeutic output accordingly through a feedback network. The precise sensing of disease signals allows for the precise production of therapeutic molecules. The therapeutic functions of synthetic devices have been demonstrated by proof-of-concept studies in mouse models of human diseases using methods such as light- or radio-wave-activated switches to regulate the precise dosage of therapeutic hormones for diabetes therapy [34,35]; synthetic gene networks to treat metabolic syndrome, obesity, and hypertension [37,38,45]; smart logic gates to kill specific cancer cells [62,63]; and synthetic circuits to reprogram immune cells [69,70,73]. These pioneering studies demonstrate the great potential of synthetic biology-based therapies to provide novel therapeutic strategies for future clinical applications [79,80].

In the past, synthetic biology has been focused on assembling prokaryotic systems, and a large proportion of the available basic biobricks have been derived from bacterial systems [26–29,31,81–85]. Recently, receptor-based methods, especially the G-protein-coupled receptors (GPCRs), inspired by synthetic biology have become a promising option for therapeutic applications. Some pioneering studies have demonstrated the great potential

of using GPCRs in synthetic gene networks for therapeutic purposes [30,34,37,38,45,78,86,87]. The utilization of human-derived proteins such as GPCRs has advantages over the heterologous proteins from bacterial systems because human-derived proteins can avoid the immune response *in vivo* and thus improve safety for clinical applications in the future.

Cell-based therapies are considered a very promising technique in the next generation of medicine [74]. With the development of synthetic biology, scientists can design and engineer versatile therapeutic microbial and mammalian cells to combat different diseases. However, nearly all the reported studies on the engineering of synthetic gene circuits in mammalian cells have been performed in cell lines, which are often utilised in mammalian synthetic biology [28,31,34,38,83,84]. The challenge is to make sure that the engineered cells containing synthetic gene networks will have no side effects in humans. Nevertheless, microencapsulation technology may become an option for the transplantation of engineered cells into the human body for therapeutic purposes [88]. Using this technology, engineered cells containing synthetic therapeutic gene circuits could be sufficiently separated from the rest of the tissue by a semipermeable (e.g., alginate–polylysine–alginate) membrane that allows free communication with the body fluids and at the same time protects the cells from the body's immune system [89]. A pioneering clinical study confirmed that cells implanted into humans inside insulating and immunoprotective microcontainers were able to coordinate expression of therapeutic proteins [90]. Another way to minimise the risk is to engineer primary cells or stem cells directly isolated from patients. Induced pluripotent stem cell (iPSC) technology provides a way to produce and engineer autologous cells from patients, even including elderly patients [91]. The advantages of engineering primary cells are obvious, including immunocompatibility and non-carcinogenicity.

It is an ambitious goal to use synthetic-biology-based biomedical devices in the clinics. To achieve this aim, it is important to standardise genetic circuit design and to create more robust and higher-order genetic circuits. In the meantime, the clinical use of synthetic therapeutic circuits will face ethical and legal issues [92]. For now, although there are no synthetic gene circuits being developed for the clinic or clinical trials, synthetic biologists should design and create more reliable genetic circuits to provide evidence in support of the great human therapeutic benefit that could arise from clinical trials of synthetic biology-based treatments and to demonstrate the advantages compared with traditional therapeutic strategies. Despite the difficulties ahead, there are increasing numbers of synthetic biology-related studies on the way that will have a deep impact on the treatment strategies of the 21st century. We believe that along with the progress that has come from proof-of-concept studies, mammalian synthetic biology will become a universally accepted tool for clinical medical practice in this century.

Acknowledgements

We thank Pratik Saxena and Mingqi Xie for generous advice and thorough comments on the manuscript. This work was supported by a European Research Council (ERC) advanced Grant (ProNet No. 321381) and by the Cantons of Basel and the Swiss Confederation within the INTERREG IV A.20 tri-national research program.

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